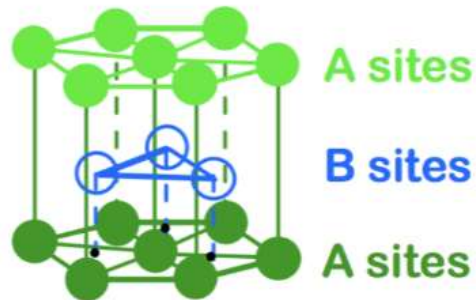
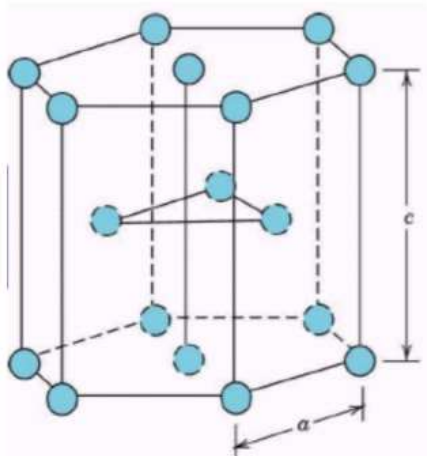


APS110 Tutorial 5

Crystal structure (quick HCP recap)

Hexagonal Close Packed (HCP)

- Coordination # 12
- Number of atoms per unit cell: $12 (\text{corner atoms}) \times \frac{1}{6} + 3 (\text{interior atoms}) + 2 (\text{face atoms}) \times \frac{1}{2} = 6 \text{ atoms / unit cell}$
- $c/a = 1.63$ For most HCP metals



Explain how FCC and HCP are the same (crystal orientation)

Explain the difference between ABABAB and ABCABC stackings

Defects

See slides

Boltzmann not the same as Maxwell-Boltzmann distribution (only one on google images)

Polymer intro

See slides

The face centered cubic and hexagonal close packed structures both have a packing factor of 0.74, consist of closely packed planes of atoms, and have a coordination number of 12. The difference between the fcc and hcp is the stacking sequence. The hcp layers cycle among the two equivalent shifted positions whereas the fcc layers cycle between three positions

Learning objectives of the 2 weeks (accessible in the syllabus section in Quercus)

Make sure you're familiar with these concepts for your tutorials

- Identify on the stress-strain curve for a typical metal the following points or regions: uniform elastic deformation, uniform plastic deformation, non-uniform plastic deformation, onset of necking - Explain why the engineering stress decreases following the onset of necking - Compare and contrast uniform and non-uniform deformation
- Demonstrate how plastic deformation in metals occurs through the movement of linear crystalline defects - Compare and contrast between elastic and plastic deformation in terms of the movement of atoms in a lattice - Show how knowledge of the mechanism of plastic deformation allows understanding of strengthening mechanisms
-Organize crystalline defects according to their dimensionality, that is, zero-dimensional, one-dimensional, two-dimensional, and three-dimensional - Explain how 0-dimensional imperfections occur and how some can increase strength of a metal - Provide several examples of 0-dimensional imperfections - Demonstrate how the vacancy population in a metal changes as a function of temperature
- Demonstrate how the distribution of particles over available energy states is characterized by the Boltzmann distribution - Show how changes in temperature are reflected in changes in the Boltzmann distribution
-Organize crystalline defects according to their dimensionality, that is, zero-dimensional, one-dimensional, two-dimensional, and three-dimensional - Explain how 1-dimensional imperfections can increase strength of a metal - Describe how the number of linear imperfections can be increased in a metal
-Organize crystalline defects according to their dimensionality, that is, zero-dimensional, one-dimensional, two-dimensional, and three-dimensional - Explain how 2-dimensional imperfections can increase the strength of a metal - Provide several examples of 2-dimensional imperfections in metals
-Organize crystalline defects according to their dimensionality, that is, zero-dimensional, one-dimensional, two-dimensional, and three-dimensional - Explain how 3-dimensional imperfections can increase the strength of a metal - Provide several examples of 3-dimensional imperfections in metals - Summarize the four major strengthening mechanisms in metals
Represent the structure of a polymer using the model of a collection of strings
Apply the string model of polymer structure to explain strengthening of polymers via chain orientation
Define the yield strength and tensile strength on the tensile stress-strain curve of a typical polymer

Questions

5. (1 point) Which of the following is true of the vacancy population in a metal?

- A. The number of vacancies decreases with an increase in temperature
- B. The number of vacancies is a measure of the porosity of a material
- C. The number of vacancies is independent of temperature
- D.** The number of vacancies increases with an increase in temperature

6. (1 point) Regarding the Boltzmann distribution, which of the following is true of a system at infinite temperature?

- A. The particles will occupy a distribution of energy states shifted towards the higher energy states
- B.** The particles will be evenly distributed across all of the possible energy states
- C. The particles will all occupy only the lowest energy state
- D. The particles will all occupy only the highest energy state

7. (1 point) The yield strength of an alloy is generally lower than that of either of the constituent pure elements.

- A. True
- B.** False

9. (1 point) Carbon dissolved in iron will be present as which of the following type(s) of crystalline imperfection?

- A. Second phase only
- B.** Point defect only
- C. Point defect and second phase
- D. Neither point defect nor second phase

13. (1 point) Plastic deformation in metals occurs through which of the following mechanisms?

- A. The simultaneous breaking of all bonds along a crystallographic plane
- B. The step-by-step movement of point defects
- C. The simultaneous breaking of all bonds along a grain boundary
- D.** The step-by-step movement of linear imperfections

14. (1 point) Which of the following best describes the distribution of particles over possible energy states (Boltzmann distribution) for a population of particles at absolute zero?

- A.** Only the lowest energy state is populated
- B. An intermediate energy state is most highly populated
- C. Only the highest energy states are populated
- D. All energy states equally populated

15. (1 point) Plastic deformation of a metal increases the yield strength for which of the following reasons?

- A. Plastic deformation reduces the number of dislocations, which increases the strength
- B.** Plastic deformation increases the number of dislocations, which inhibits dislocation movement
- C. Plastic deformation increases the grain size, which increases the strength
- D. Plastic deformation increases the number of vacancies which inhibits dislocation movement

Long answers

1) A metallic sample has a vacancy defect formation energy of 3.5 eV. The amount of vacancies increases by a factor of 100 when the sample is heated up (T_1 to T_2). If the initial temperature is 800°C, what is the final temperature?

$$N_v = N \exp\left(-\frac{Q_v}{k_B T_1}\right) \quad (1)$$

$$100 N_v = N \exp\left(-\frac{Q_v}{k_B T_2}\right) \quad (2)$$

$$\frac{(2)}{(1)} \Rightarrow 100 = \frac{\exp\left(-\frac{Q_v}{k_B T_2}\right)}{\exp\left(-\frac{Q_v}{k_B T_1}\right)} \Rightarrow \exp\left(\frac{-Q_v}{k_B T_2} + \frac{Q_v}{k_B T_1}\right)$$

$$\ln(100) = \frac{Q_v}{k_B T_1} - \frac{Q_v}{k_B T_2}$$

$$\frac{k_B}{Q_v} \ln(100) = \frac{1}{T_1} - \frac{1}{T_2}$$

$$T_2 = \left(\frac{1}{T_1} - \frac{k_B}{Q_v} \ln(100) \right)^{-1}$$

Plug in values $\Rightarrow$

$$T_1 = 800^\circ\text{C} = 1073 \text{ K}$$

$$Q_v = 3.5 \text{ eV} = 5.607 \cdot 10^{-19} \text{ J}$$

$$k_B = 1.38 \cdot 10^{-23} \text{ J/K}$$

$$T_2 = 1221.56 \text{ K}$$

$$\approx 948.6^\circ\text{C}$$

4. Match each of the following imperfections with its dimensionality by writing 0, 1, 2, or 3 in each box, for zero-dimensional, one-dimensional, two-dimensional, and three-dimensional, respectively. (1 mark each)

a. Salt dissolved in hot water

0

b. Carbon dissolved in iron

0

c. The imperfection illustrated here:



1

d. An air bubble within a piece of glass

3

e. The surface of a piece of aluminum

2

f. A grain boundary

2

g. An atom missing from its lattice site

0

h. A potassium atom occupying a lattice site normally occupied by sodium

0

i. A lump of graphite within a sample of iron

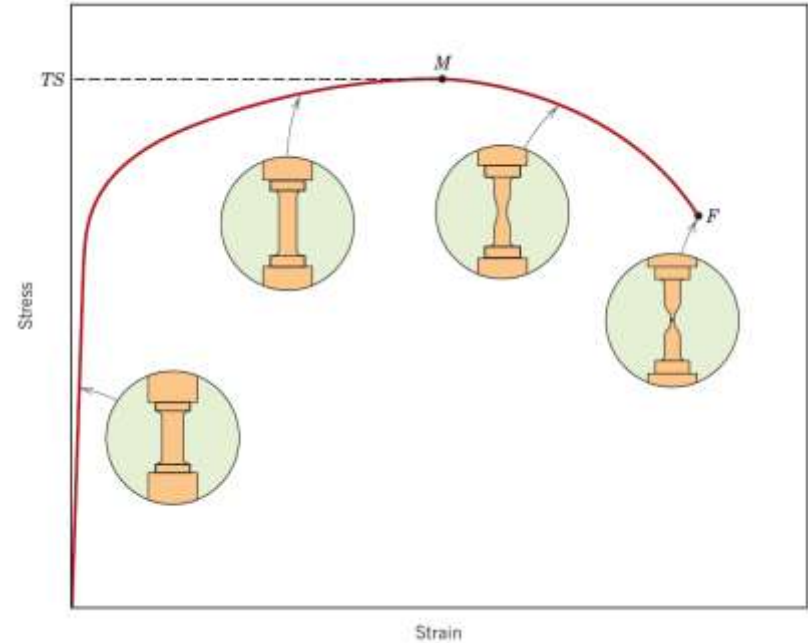
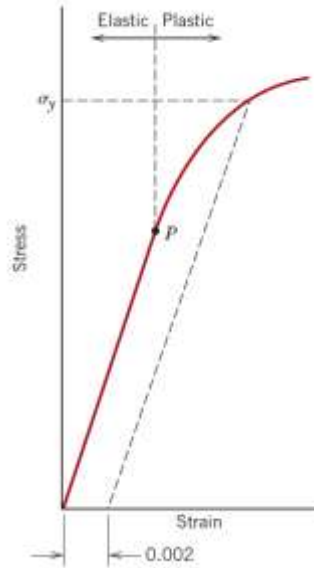
3

j. A dislocation

1

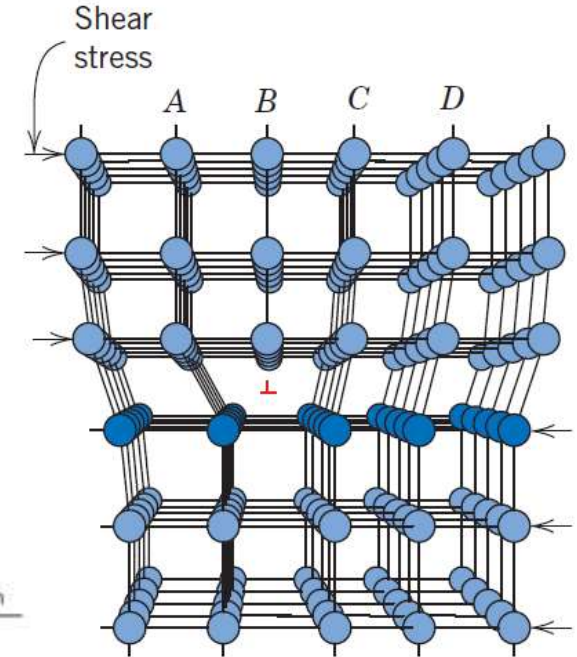
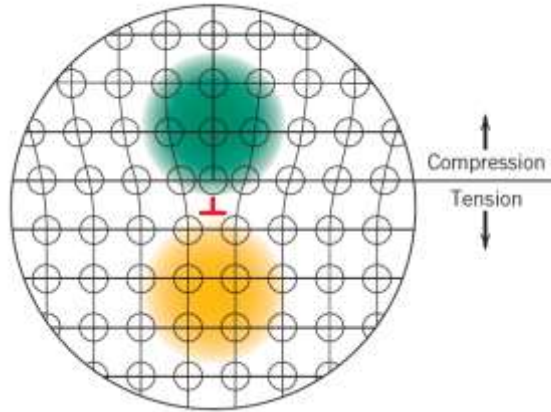
Today's Lecture Recaps

- 0.2% Offset - a recognized convention for determining yield strength
- Portional Limit
- Ultimate Tensile Stress (σ_{UTS})
- Decrease in engineering stress after σ_{UTS}
- Linear/nonlinear (Localized) elastic/plastic deformation



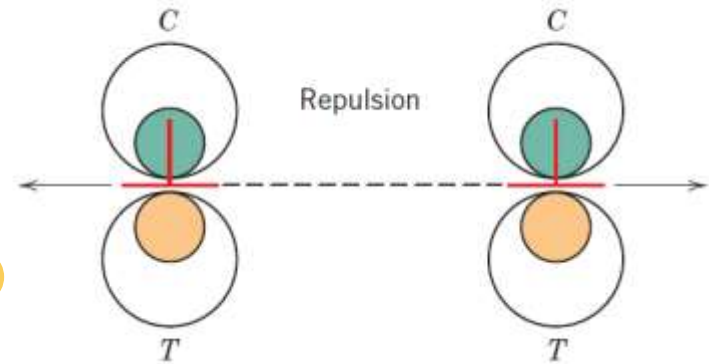
Today's Lecture Recaps

- 1D Imperfections -> Dislocations
 - Mechanisms for plastic deformation of metals
- Dislocation Density - $10^3 - 10^{10}$ dislocations /mm²
 - Carefully heat treated (annealed) to heavily deformed metals
- Lattice Strain / Strain Fields



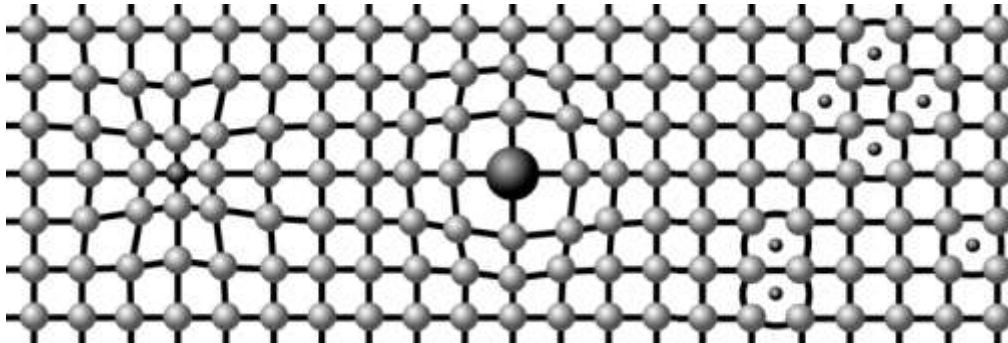
Today's Lecture Recaps

- Strain Hardening
 - Phenomenon by which a ductile metal becomes stronger and harder as it's plastically deformed
 - As more dislocations are introduced by plastic deformation, strain around dislocations interact and inhibit dislocation movement (repulsive strain field) - hence the increased resistance to plastic deformation i.e. strengthening of the material



Today's Lecture Recaps

- 0 D imperfections -> Point Defects/Vacancies
- Substitutional Impurity
 - Similar in size, crystal structure and electronegativity
- **Interstitial Impurity**
 - **If left long enough, the atoms tends to diffuse toward dislocation lines**
 - **“pin the dislocations”**

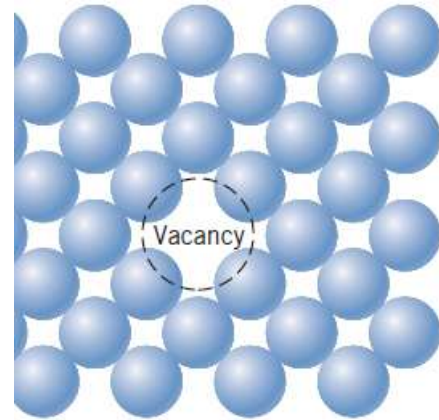


Today's Lecture Recaps

- Vacancies - missing atom
 - All crystalline solids contain vacancies
 - Thermodynamically introduced

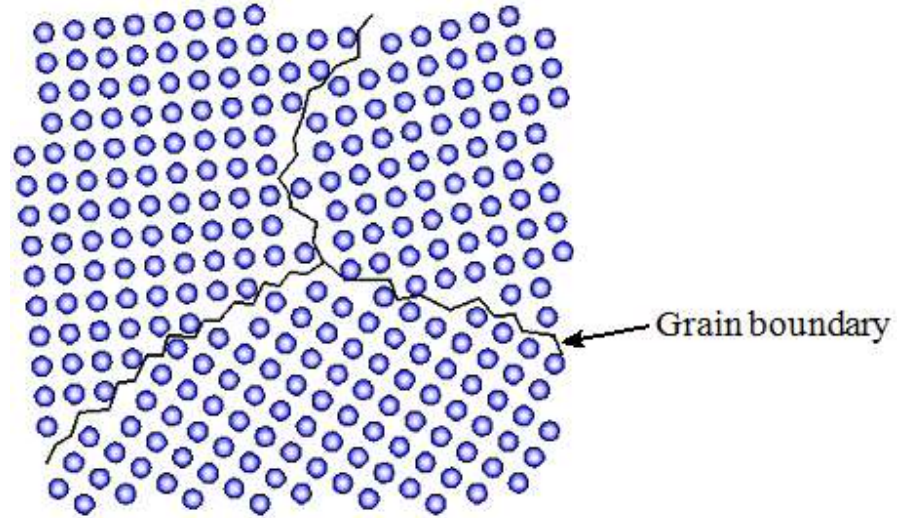
$$N_v = N \exp \left(-\frac{Q_v}{kT} \right)$$

- N = total # of atomic sites
- Q_v = energy required for the formation of a vacancy
- T = absolute temperature in K
- k = Boltzmann's constant = 1.38×10^{-23} J/atom K
- Boltzmann's distribution



Today's Lecture Recaps

- 2 D imperfections -> Grain Boundaries (GB)
- Dislocations are difficult to cross GB
 - Change in lattice direction
 - Planar mis-match/misorientation
 - Varied interatomic spacing
- Grain size reduction
 - More GBs
 - Increased grain surface energy
 - Strengthening



- What happens if you heat up high grain energy material?
 - Grain growth - High surface energy to low surface energy relaxation

Today's Lecture Recaps



- 3D Imperfections - Volume Defects / Second Phase Particles
 - Ex. Fe_3C in steel
 - If you put a little bit of C in Fe - solid solution - 0 D (substitutional impurities)
 - If you have a lot - react and form Fe_3C
- Similar to GB but usually exhibit hard and brittle behavior
 - Difficult for dislocations to cross
 - Strengthening of the material

Today's Lecture Recaps

- Polymer Mechanical Behaviors
 - Continues to support load after necking
 - For amorphous material, while neck elongates - the long polymer chains start to line up - stronger modulus

